Grade 8
Unit 8
Lesson 2 Practice

Name__Answer Key_______
Date___________________
Period__________________

Ms. Aguirre was buying notebooks and pencils. A box of notebooks cost $\$11$ more than a box of pencils. She bought 6 boxes of each and spent $\$162$ altogether. The equation $6(p + p + 11) = 162$ models the situation, where p is the cost of one box of pencils.

Landry, Chiara, and Roy were all solving the equation to help Ms. Aguirre. Landry's first step was to divide both sides by 6. Chiara started solving the equation by using the distributive property. Roy started solving the equation by combining like terms. Complete each person's method.

1. Landry's Method	**2. Chiara's Method**	**3. Roy's Method**
$6(p + p + 11) = 162$ $\frac{6(p + p + 11)}{6} = \frac{162}{6}$ $p + p + 11 = 27$ $2p + 11 = 27$ $2p = 16$ $p = 8$	$6(p + p + 11) = 162$ $6p + 6p + 66 = 162$ $12p + 66 = 162$ $-66 \quad -66$ $12p = 96$ $p = 8$	$6(p + p + 11) = 162$ $6(2p + 11) = 162$ $2p + 11 = 27$ $2p = 16$ $p = 8$

4. Explain which method you prefer for solving this equation.

Answers will vary.

5. Complete each statement to interpret the solution in this context.

Each box of pencils cost $__8___.

Each box of notebooks cost $__19___.

6. Horatio has the same number of quarters as dimes. He also has 3 nickels. If those are the only types of coins he has and he has a total of $\$1.90$, the situation can be represented by the equation $0.25q + 0.10q + 3(0.05) = 1.90$, where q is the number of quarters Horatio has. Solve the equation to determine the number of quarters he has.

$0.25q + 0.10q + 3(0.05) = 1.90$
$0.35q + 0.15 = 1.90$
$0.35q = 1.75$
$q = 5$ quarters

7. Amaya is on her school's basketball team. In the last game, she made 4 two-point shots, some three-point shots, and 7 one-point free throws. She scored a total of 30 points. Use the equation $2(4) + 3p + 1(7) = 30$ to find p, the number of three-point shots Amaya made.

$8 + 3p + 7 = 30$

$3p + 15 = 30$

$3p = 15$

$p = 5$

8. Sylvie had a coupon for $\$1.75$ off each movie ticket. She bought 4 tickets for her and her friends and also spent $\$18.89$ on concessions. Her total was $\$52.09$. Use the equation $4(t - 1.75) + 18.89 = 52.09$ to determine t, the original cost of each ticket.

$4t - 7 + 18.89 = 52.09$

$4t - 7 = 33.20$

$4t = 40.2$

$t = \$10.05$

9. Soren has a job as a mechanic at an auto-body shop. He is paid a rate of $\$140$ per day. He earns an additional $\$21.40$ for every hour he works overtime. On Saturday, the shop was busy, and Soren worked overtime and earned a total of $\$214.90$. Use the equation $140 + 21.40h = 214.90$ to find h, the number of hours Soren worked overtime.

$140 + 21.40h = 214.90$

$21.40h = 74.90$

$h = 3.5$ hours

10. Six friends went to a carnival. They each bought an admissions ticket, a ride package for $\$9.75$, and a cotton candy for $\$3.80$. Their total was $\$153.30$. Use the equation $6(a + 9.75 + 3.80) = 153.30$ to determine a, the cost of admissions.

$6a + 58.5 + 22.8 = 153.30$

$6a + 81.3 = 153.30$

$6a = 72$

$a = \$12$